Chemistry Tech 9 — Solved Exercises (PCTB)

Class 9 — Solved Exercises

Punjab Curriculum and Textbook Board (PCTB)

12 chapters — 258 exercises

Chapter 1: States of Matter and Phase Changes

1. Tick the correct answer.

(i) Hazardous effects of shopping bags are studied in:

(a) Geochemistry

(b) Inorganic Chemistry

(c) Analytical Chemistry

(d) Environmental Chemistry ✓

Correct answer: Environmental Chemistry

1. Tick the correct answer.

(ii) The man-made polymer is:

(a) Starch

(b) Polystyrene ✓

(c) Protein

(d) Cellulose

Correct answer: Polystyrene

1. Tick the correct answer.

(iii) The crystals of which substance has rhombic shape?

(a) Brass

(b) Sulphur ✓

(c) Graphite

(d) Bronze

Correct answer: Sulphur

1. Tick the correct answer.

(iv) Which liquid among the following is a colloidal solution?

(a) Milk ✓

(b) Slaked lime used for white wash

(c) Vinegar solution

(d) Mixture of AgCl in water

Correct answer: Milk

1. Tick the correct answer.

(v) Which of the following is a heterogeneous mixture?

(a) A solution of calcium hydroxide in water

(b) A solution of potassium nitrate in water

(c) Hot chocolate

(d) Concrete mixture ✓

Correct answer: Concrete mixture

1. Tick the correct answer.

(vi) When the tiny visible particles of a substance are dispersed through a medium, the mixture is named as:

(a) True solution

(b) Colloid

(c) Suspension ✓

(d) Saturated solution

Correct answer: Suspension

1. Tick the correct answer.

(vii) A solution of KClO3 has a solubility of about 13.2 g per 100 cm3 at 40 C. How its solubility will be affected, if you decrease the temperature?

(a) The solubility will increase

(b) The solubility will decrease ✓

(c) The solubility will remain the same

(d) The solubility will first increase with temperature and then it will decrease

Correct answer: The solubility will decrease

1. Tick the correct answer.

(viii) You are studying the rate of hydrolysis of starch under different conditions of temperature. In which branch of chemistry this topic will fall?

(a) Organic Chemistry

(b) Analytical Chemistry

(c) Biochemistry

(d) Physical Chemistry ✓

Correct answer: Physical Chemistry

Note on Question 1 (viii)

Note: Why is the rate of starch hydrolysis Physical Chemistry and not Biochemistry?

The chapter itself defines Physical Chemistry as the branch used to predict and change the rates of reactions. This question asks about the RATE of a reaction under different temperatures, so it is a rates study and falls in Physical Chemistry. Biochemistry would study the life processes and molecules themselves, not the measurement of the reaction rate.

2. Questions for Short Answers

i. Why is there a need to divide Chemistry into many branches? Give three reasons.

First, chemistry covers a very wide and complex subject, so no one person can study all of it in depth. Second, dividing it lets scientists concentrate on one specific area and make faster progress in it. Third, each branch develops its own methods and instruments suited to its own problems, which leads to new

discoveries and applications.

2. Questions for Short Answers

ii. Reactions may take place due to electrons present outside the nucleus or they may take place inside the nucleus. Which branches of Chemistry cover these two types of reactions?

Reactions involving electrons outside the nucleus are ordinary chemical reactions, studied by the main branches of chemistry such as Physical, Inorganic and Organic Chemistry. Reactions taking place inside the nucleus are studied by Nuclear Chemistry, which deals with radioactivity and changes in the nuclei of atoms.

2. Questions for Short Answers

iii. What types of problems are solved in analytical chemistry?

Analytical chemistry solves problems of separation, identification and measurement: what substances are present in a sample, and in what amount or concentration. It uses modern instruments to analyse materials such as water, food, medicines, minerals and industrial products.

2. Questions for Short Answers

iv. Both graphite and graphene have hexagonal layered structures. What is the difference?

Graphite is made of many hexagonal layers of carbon atoms stacked on top of one another. Graphene is a single one-atom-thick hexagonal layer of carbon atoms. In short, graphene is one layer, while graphite is a stack of many such layers.

2. Questions for Short Answers

v. What is the importance of graphene?

Graphene is the thinnest and one of the strongest materials known, yet it is light and flexible. It conducts electricity and heat better than most materials and is almost transparent. Because of this it is used in electronics, batteries, sensors, strong composite materials and medical devices.

2. Questions for Short Answers

vi. Which form of matter do most of the material things in this world belong to?

Most material things belong to mixtures. Very few things in daily life are pure elements or pure compounds. Air, water from natural sources, soil, rocks, food, milk and alloys are all mixtures of two or more substances mixed in any ratio.

3. Constructed Response Questions

i. Give the reason of brilliance shown by diamond. Can you improve it?

Diamond is brilliant because it has a very high refractive index (2.42) and it bends light strongly, so light entering the stone is totally internally reflected many times and then dispersed into colours before it leaves. Yes, its brilliance can be improved by cutting it into the right shape, such as the brilliant cut, with well-placed facets, and by polishing its surfaces perfectly so that the maximum light is reflected back to the eye.

3. Constructed Response Questions

ii. Why do different compounds have different solubilities in water at a particular temperature?

Solubility depends on how strongly the particles of the solute attract one another compared with how strongly they attract water molecules. Compounds whose particles are held together very strongly, or which attract water only weakly, dissolve little. Compounds that form strong attractions with water dissolve more.

Temperature, pressure and the nature of both solute and solvent also affect the result.

3. Constructed Response Questions

iii. Why NaCl can not be crystallized from water just like KNO3?

Crystallisation by cooling works only when solubility falls sharply as temperature falls. The solubility of KNO3 rises steeply with temperature, so cooling a hot saturated solution deposits a large crop of crystals. The solubility of NaCl changes very little with temperature, so cooling deposits almost no crystals. Salt is therefore recovered by evaporating the water, not by cooling.

3. Constructed Response Questions

iv. Why graphite is slippery to touch? Which property of graphite enables it to be used as lubricant?

Graphite is slippery because it is built of flat layers of carbon atoms, and the layers are held together only by weak forces. The layers slide easily over one another when touched or rubbed. This same property, the easy sliding of its layers, together with its soft and greasy feel, makes graphite a good dry lubricant.

4. Descriptive Questions

i. Mention the name of the branch of Chemistry in which you will study each of the following topics. (a) Rate of a reaction (b) Digestion of food in human body (c) Properties of colloidal solution (d) Ecosystem (e) Reactions taking place during fireworks (f) Measurement of the absorption of wavelength with the help of ultraviolet spectrometer.

(a) Rate of a reaction: Physical Chemistry, which studies reaction rates. (b) Digestion of food in human body: Biochemistry, which studies chemical

processes in living organisms. (c) Properties of colloidal solution: Physical Chemistry, which studies the behaviour of matter at molecular level. (d) Ecosystem: Environmental Chemistry, which studies chemical phenomena in air, soil and water. (e) Reactions taking place during fireworks: Physical Chemistry, which studies the energy and rates of rapid reactions. (f) Measurement of absorption of wavelength with ultraviolet spectrometer: Analytical Chemistry, which uses instruments to analyse substances.

4. Descriptive Questions

ii. What are allotropic forms? Explain the allotropic forms of carbon and sulphur. How does coal differ from diamond?

Allotropic forms are different physical forms of the same element in the same state, with different arrangements of atoms and different properties. Carbon exists as diamond, in which each carbon atom is bonded to four others in a hard three-dimensional network; as graphite, in layers of hexagonal rings; and as fullerene, in closed cage-like molecules. Sulphur exists as rhombic sulphur, stable below 96 C, and monoclinic sulphur, stable above 96 C, both made of S8 rings but packed differently. Coal differs from diamond because coal is impure, soft, black and amorphous carbon mixed with other substances, while diamond is pure, hard, transparent, crystalline carbon.

4. Descriptive Questions

iii. Define solubility of a solute. How does the solubility of solutes change with the increase in temperature?

Solubility is the maximum amount of a solute, in grams, that dissolves in 100 g (or 100 cm3) of a solvent at a given temperature to form a saturated solution. For most solid solutes, solubility increases with temperature, for example copper sulphate, sodium nitrate and potassium nitrate dissolve more in hot water. A few

solids behave oppositely, for example the solubility of calcium hydroxide decreases as temperature increases. For gases, solubility in water decreases as temperature increases.

4. Descriptive Questions

iv. Differentiate between the areas which are studied under inorganic and organic chemistry.

Inorganic chemistry studies elements and compounds that contain little or no carbon: metals, non-metals, salts, acids, bases and minerals, their synthesis, composition, properties and structure. Organic chemistry studies carbon compounds, hydrocarbons and their derivatives, their structure, formation, properties, composition and reactions. Organic compounds are found in all forms of life and are essential for it, while inorganic compounds include fertilisers, medicines, catalysts, pigments and coatings.

Chapter 2: Atomic Structure

In-chapter Exercise (book p. 17)

How would you compare the masses of the atoms of Mg and Cl?

By their relative atomic masses. Magnesium has a relative atomic mass of about 24 and chlorine of about 35.5, so one atom of chlorine is about one and a half times heavier than one atom of magnesium. For equal numbers of atoms, a sample of chlorine is heavier than a sample of magnesium in the ratio 35.5 to 24.

1. Tick the correct answer.

(i) How many electrons can be accommodated at the most in the third shell of the elements?

(a) 8

(b) 18 ✓

(c) 10

(d) 32

Correct answer: 18

1. Tick the correct answer.

(ii) What information was obtained from discharge tube experiments?

(a) Structure of atom was discovered.

(b) Neutrons and protons were discovered.

(c) Electrons and protons were discovered. ✓

(d) Presence of nucleus in an atom was discovered.

Correct answer: Electrons and protons were discovered.

1. Tick the correct answer.

(iii) What does keep the particles present in the nucleus intact?

(a) Particles are held together by strong nuclear force.

(b) Particles are held together by weak nuclear force.

(c) Particles are held together by electrostatic force.

(d) Particles are held together by dipolar force.

Correct answer: Particles are held together by strong nuclear force.

1. Tick the correct answer.

(iv) How do electrons keep themselves away from the oppositely charged nucleus?

(a) By keeping themselves stationary

(b) By revolving around the nucleus

(c) Due to their wave-like nature

(d) A magnetic field around the nucleus keeps them away

Correct answer: By revolving around the nucleus

2. Questions for Short Answers

i. Why is it said that almost all the mass of an atom is concentrated in its nucleus?

The nucleus contains the protons and neutrons, which are about 1836 times heavier than an electron. The electrons outside the nucleus have almost no mass compared with them. Therefore, although the nucleus is tiny in size, it holds practically the whole mass of the atom.

2. Questions for Short Answers

ii. Why are elements different from one another?

Elements are different because they have different numbers of protons, that is, different atomic numbers. The number of protons fixes the number of electrons in a neutral atom, and the electronic arrangement fixes the chemical properties.

Hence each element behaves differently from every other element.

2. Questions for Short Answers

iii. How many neutrons are present in 210 over 83 Bi?

Mass number = 210, atomic number = 83. Number of neutrons = mass number minus atomic number = 210 minus 83 = 127. So 127 neutrons are present in this atom of bismuth.

2. Questions for Short Answers

iv. How can an atom absorb and evolve energy?

An atom absorbs energy when one of its electrons jumps from a lower energy shell to a higher energy shell. An atom evolves energy when an excited electron falls back from a higher shell to a lower shell, and the energy is given out, usually as light of a definite wavelength.

3. Constructed Response Questions

i. Why does the energy of electron increase as we move from first shell to second shell?

The first shell lies closest to the nucleus, so its electrons are held most strongly and have the lowest energy. The second shell is farther from the nucleus, so its electrons are held less strongly and possess higher energy. Energy of the shells therefore increases as we move away from the nucleus.

3. Constructed Response Questions

ii. Why is it needed to lower the pressure of the gas inside the discharge tube?

At normal pressure the gas molecules are crowded, and the electrons knocked out of them collide with molecules again and again, so no steady stream reaches the other end. When the pressure is lowered, the molecules become few and far

apart, so the electrons travel freely across the tube and the cathode rays can be observed.

3. Constructed Response Questions

iii. During discharge tube experiments, how did the scientists conclude that the same type of electrons and protons are present in all the elements?

The cathode rays were found to have the same charge-to-mass ratio no matter which gas was used in the tube and no matter which metal the electrodes were made of. Similarly, the positively charged canal rays showed the same kind of particles from different gases. Since the particles did not depend on the material used, scientists concluded that electrons, and likewise protons, are the same in all elements.

4. Descriptive Questions

i. Describe how electron and proton were discovered?

The electron was discovered through discharge tube experiments. When a high voltage was applied across a gas at low pressure, rays travelled from the cathode to the anode. These cathode rays were bent by electric and magnetic fields, showing they were negatively charged particles, and Thomson measured their charge-to-mass ratio and named them electrons. The proton was discovered through canal rays: Goldstein found positively charged rays travelling in the opposite direction through holes in the cathode. Rutherford later showed that the lightest of these positive particles, from hydrogen, is a fundamental particle of all nuclei, and it was named the proton.

4. Descriptive Questions

ii. Explain the structure of a hydrogen atom.

A hydrogen atom is the simplest atom. Its nucleus contains a single proton and

no neutron. A single electron revolves around this nucleus in the first shell. Its atomic number is 1 and its mass number is 1, and its electronic configuration is 1.

4. Descriptive Questions

iii. Explain the concepts of atomic number, mass number and relative atomic mass.

Atomic number is the number of protons in the nucleus of an atom, and it fixes the identity of the element. Mass number is the total number of protons and neutrons in the nucleus. Relative atomic mass is the average mass of the atoms of an element compared with one-twelfth of the mass of a carbon-12 atom; it is expressed in atomic mass units (amu), where 1 amu equals one-twelfth of the mass of a carbon-12 atom.

Chapter 3: Chemical Bonding

1. Tick the correct answer.

(i) When molten copper and molten zinc are mixed together, they give rise to a new substance called brass. Predict what type of bond is formed between copper and zinc.

(a) Coordinate covalent bond

(b) Ionic bond

(c) Metallic bond ✓

(d) Covalent bond

Correct answer: Metallic bond

1. Tick the correct answer.

(ii) Which element is capable of forming all the three types of bonds; covalent, coordinate covalent and ionic?

(a) Carbon

(b) Oxygen ✓

(c) Magnesium

(d) Silicon

Correct answer: Oxygen

1. Tick the correct answer.

(iii) Which of the following bonds is expected to be the weakest?

(a) C-C

(b) Cl-Cl

(c) O-O

(d) F-F ✓

Correct answer: F-F

Note on Question 1 (iii)

Note: Why is F-F marked as the weakest bond?

Fluorine atoms are the smallest in the list and each carries three lone pairs of electrons crowded close to the bond, so the lone pairs repel one another strongly and weaken the F-F bond. This is the expected answer from the small-size and lone-pair repulsion reasoning taught at this level. Note for the careful reader: by measured bond energies the O-O bond (about 146 kJ/mol) is marginally weaker than the F-F bond (about 155 kJ/mol); the chapter does not give a bond-energy table, so the expected textbook answer, F-F, is marked.

1. Tick the correct answer.

(iv) Which form of carbon is used as a lubricant?

(a) Coal

(b) Diamond

(c) Graphite 

(d) Charcoal

Correct answer: Graphite

1. Tick the correct answer.

(v) Which metal has the lowest melting point?

(a) Li

(b) Na

(c) K

(d) Rb 

Correct answer: Rb

1. Tick the correct answer.

(vi) Which ionic compound has the highest melting point?

(a) NaCl ✓

(b) KCl

(c) LiCl

(d) RbCl

Correct answer: NaCl

1. Tick the correct answer.

(vii) Which compound contains both covalent and ionic bonds?

(a) MgCl2

(b) NH4Cl ✓

(c) CaO

(d) PCl3

Correct answer: NH4Cl

1. Tick the correct answer.

(viii) Which among the following has a double covalent bond?

(a) Ethane

(b) Methane

(c) Ethylene ✓

(d) Acetylene

Correct answer: Ethylene

2. Questions for Short Answers

i. What type of elements lose their outer electron easily and what type of elements gain electron easily?

Metals lose their outer electrons easily because they have low ionisation energies and their nuclei hold the outer electrons weakly; they are electropositive. Non-metals gain electrons easily because they have high

electron affinities and nearly complete outer shells; they are electronegative.

2. Questions for Short Answers

ii. Why do lower molecular mass covalent compounds exist as gases or low boiling liquids.

In small covalent molecules the forces between the molecules are very weak. Only a small amount of heat energy is needed to overcome these weak intermolecular forces and separate the molecules. They therefore boil at low temperatures and exist as gases or low boiling liquids at room temperature.

2. Questions for Short Answers

iii. Give one example of an element which exists as a crystalline solid and it has covalent bonds between its atoms.

Diamond is such an element. It is a crystalline solid in which every carbon atom is joined to four other carbon atoms by strong covalent bonds in a three-dimensional network, which is why diamond is extremely hard.

2. Questions for Short Answers

iv. Which property of metals makes them malleable and ductile?

The metallic bond makes metals malleable and ductile. The positive metal ions sit in a sea of mobile electrons, and when a force is applied, one layer of ions slips over another without breaking the bond, because the mobile electrons at once hold the new positions together. Metals can therefore be beaten into sheets and drawn into wires.

2. Questions for Short Answers

v. Is coordinate covalent bond a strong bond?

Yes. Once a coordinate covalent bond is formed, it is just as strong as an

ordinary covalent bond, because in both cases the two atoms are held together by a shared pair of electrons. The only difference is the origin of the pair: in a coordinate bond both electrons are donated by one atom, as in the ammonium ion and the hydroxonium ion.

2. Questions for Short Answers

vi. Write down dot and cross formula of HNO3.

In nitric acid the hydrogen atom is joined by a single bond to an oxygen atom, and that oxygen is joined by a single bond to the nitrogen atom. The nitrogen atom is joined by a double bond to a second oxygen atom and by a coordinate bond to a third oxygen atom. In the dot and cross picture the shared pairs are shown as dots for one atom and crosses for the other; the nitrogen carries a positive charge and the singly bonded oxygen carries a negative charge, so the molecule as a whole is neutral.

3. Constructed Response Questions

i. Why HF is a liquid while HCl is a gas?

Hydrogen fluoride molecules are held together by strong hydrogen bonds, because fluorine is very small and highly electronegative and attracts the hydrogen of a neighbouring molecule strongly. A large amount of heat is needed to break these bonds, so HF boils at 19.5 C and is a liquid at room temperature. Hydrogen chloride molecules are held only by weak dipole forces, so HCl boils at a much lower temperature and is a gas.

3. Constructed Response Questions

ii. Why covalent compounds are generally not soluble in water?

Water is a polar solvent, and it dissolves substances best when it can attract their particles strongly. Most covalent compounds are non-polar, so water

molecules attract them only weakly, and this weak attraction cannot overcome the forces holding the covalent substance together. Polar covalent compounds such as hydrogen chloride and ammonia are exceptions and do dissolve in water.

3. Constructed Response Questions

iii. Why does iodine exist as a solid while Cl2 exist as a gas?

Iodine molecules are much larger and contain many more electrons than chlorine molecules, so the London dispersion forces between iodine molecules are far stronger. Much more heat energy is needed to separate iodine molecules, so iodine is a solid at room temperature, while chlorine, with weak forces between its small molecules, is a gas.

4. Descriptive Questions

i. Explain the formation of an ionic bond and a covalent bond.

An ionic bond is formed by the complete transfer of one or more electrons from a metal atom to a non-metal atom. For example, a sodium atom gives its outer electron to a chlorine atom; both ions then have the electronic configuration of a noble gas, and the strong electrostatic attraction between the positive sodium ion and the negative chloride ion is the ionic bond. A covalent bond is formed by the mutual sharing of electron pairs between two non-metal atoms, so that both atoms complete their outer shells, for example two hydrogen atoms share one pair to form H2, and carbon shares four pairs with four hydrogen atoms to form methane.

4. Descriptive Questions

ii. How do ions arrange themselves to form NaCl crystal.

In a sodium chloride crystal the sodium ions and chloride ions arrange

themselves in a regular three-dimensional pattern called a face-centred cubic lattice. Each sodium ion is surrounded by six chloride ions, and each chloride ion is surrounded by six sodium ions. This regular packing goes on in all directions, giving a hard, cubical crystal with a high melting point.

4. Descriptive Questions

iii. Explain the properties of metals keeping in view the nature of metallic bond.

In a metallic bond the metal atoms release their outer electrons, which move freely through the whole piece of metal as a sea of electrons holding the positive ions together. The free electrons make metals good conductors of heat and electricity and give them their shine, because they reflect light. When a force is applied, layers of ions slip over one another without breaking the bond, so metals are malleable and ductile. The strong attraction between the ions and the electron sea gives metals high melting and boiling points, and the close packing of the ions makes them hard and heavy.

4. Descriptive Questions

iv. Compare the properties of ionic and covalent compounds.

Ionic compounds are usually hard crystalline solids with high melting and boiling points, because strong electrostatic forces hold their ions together. They conduct electricity when molten or dissolved in water, but not as solids, and they are usually soluble in water. Covalent compounds are usually gases, liquids or soft solids with low melting and boiling points, because only weak forces hold their molecules together. They do not conduct electricity in any state, and they are usually soluble in organic solvents rather than in water.

4. Descriptive Questions

v. How will you explain the electrical conductivity of graphite crystals?

In graphite each carbon atom is bonded to only three other carbon atoms, so each atom has one electron left over that is not used in bonding. These extra electrons are delocalised, that is, they are free to move along the flat layers of carbon atoms. When a voltage is applied, these mobile electrons flow through the layers, so graphite conducts electricity, unlike diamond, in which all four electrons of every carbon atom are locked in bonds.

4. Descriptive Questions

vi. Why are metals usually hard and heavy?

Metals are hard because the strong metallic bond holds their positive ions firmly in a regular lattice, and a large force is needed to disturb this arrangement. Metals are heavy because their atoms are closely packed together in the lattice and most metal atoms have fairly large masses, so a small volume of metal contains a large mass, giving a high density.

Chapter 4: Stoichiometry

1. Tick the correct answer.

(i) How many atoms are present in one gram of H2O?

(a) 1002 x 10^23 atoms ✓

(b) 6.022 x 10^23 atoms

(c) 0.334 x 10^23 atoms

(d) 2.004 x 10^23 atoms

Correct answer: 1002 x 10^23 atoms

Note on Question 1 (i)

Note: The book prints option (a) as 1002 x 10^23, and the working shows what is meant.

Step 1: Molar mass of H2O = 18 g/mol. Step 2: Moles in 1 g = 1 / 18 = 0.0556 mol. Step 3: Number of molecules = 0.0556 x 6.022 x 10^23 = 0.334 x 10^23 molecules. Step 4: Each H2O molecule has 3 atoms, so atoms = 3 x 0.334 x 10^23 = 1.002 x 10^23 atoms. Option (a) is therefore the correct choice, read as 1.002 x 10^23 atoms; the decimal point is missing in the printed option. Option (c), 0.334 x 10^23, is the number of molecules, not atoms.

1. Tick the correct answer.

(ii) Which is the correct formula of calcium phosphide?

(a) CaP

(b) CaP2

(c) Ca2P3

(d) Ca3P2 ✓

Correct answer: Ca3P2

1. Tick the correct answer.

(iii) How many atomic mass units (amu) are there in one gram?

(a) 1 amu

(b) 10^23 amu

(c) 6.022 x 10^23 amu

(d) 6.022 x 10^22 amu

Correct answer: 6.022 x 10^23 amu

1. Tick the correct answer.

(iv) Structural formula of 2-hexene is CH3-CH=CH-(CH2)2-CH3. What will be its empirical formula?

(a) C2H2

(b) CH

(c) C6H12

(d) CH2

Correct answer: CH2

1. Tick the correct answer.

(v) How many moles are there in 25 g of H2SO4?

(a) 0.765 moles

(b) 0.51 moles

(c) 0.255 moles

(d) 0.4 moles

Correct answer: 0.255 moles

1. Tick the correct answer.

(vi) A necklace has 6 g of diamonds in it. What are the number of carbon atoms in it?

(a) 6.02 x 10^23

(b) 12.04 x 10^23

(c) 1.003 x 10^23

(d) 3.01 x 10^23 ✓

Correct answer: 3.01 x 10^23

1. Tick the correct answer.

(vii) What is the mass of Al in 204 g of aluminium oxide, Al2O3?

(a) 26 g

(b) 27 g

(c) 54 g

(d) 108 g ✓

Correct answer: 108 g

1. Tick the correct answer.

(viii) Which one of the following compounds will have the highest percentage of the mass of nitrogen?

(a) CO(NH2)2

(b) N2H4 ✓

(c) NH3

(d) NH2OH

Correct answer: N2H4

1. Tick the correct answer.

(ix) When one mole of each of the following compounds is reacted with oxygen, which will produce the maximum amount of CO2?

(a) Carbon

(b) Diamond

(c) Ethane (C2H6) ✓

(d) Methane (CH4)

Correct answer: Ethane (C2H6)

1. Tick the correct answer.

(x) What mass of 95% CaCO3 will be required to neutralize 50 cm3 of 0.5 M HCl solution?

(a) 9.5 g

(b) 1.25 g

(c) 1.32 g

(d) 1.45 g

Correct answer: 1.32 g

Working for the numerical MCQs

Working: Questions 1 (v), (vi), (vii), (viii) and (x) in full steps.

(v) Molar mass of H2SO4 = 98 g/mol. Moles = 25 / 98 = 0.255 mol. (vi) Moles of carbon = 6 / 12 = 0.5 mol. Atoms = 0.5 x 6.022 x 10^23 = 3.01 x 10^23. (vii) Molar mass of Al2O3 = 102 g/mol. Moles in 204 g = 204 / 102 = 2 mol. Each mole of Al2O3 has 2 mol of Al, so moles of Al = 4 mol. Mass of Al = 4 x 27 = 108 g. (viii) Percentage of N: urea CO(NH2)2 = 28/60 = 46.7 percent; N2H4 = 28/32 = 87.5 percent; NH3 = 14/17 = 82.4 percent; NH2OH = 14/33 = 42.4 percent. N2H4 is highest. (x) Moles of HCl = 0.5 x 50/1000 = 0.025 mol. Equation: CaCO3 + 2HCl gives CaCl2 + H2O + CO2, so moles of CaCO3 = 0.025 / 2 = 0.0125 mol. Pure mass = 0.0125 x 100 = 1.25 g. Impure mass needed = 1.25 x 100 / 95 = 1.32 g.

2. Questions for Short Answers

i. Write down the chemical formula of barium nitride.

Barium forms Ba2+ ions and nitrogen forms N3- ions. To balance the charges,

three Ba2+ ions (total +6) combine with two N3- ions (total -6). The formula is Ba3N2.

2. Questions for Short Answers

ii. Find out the molecular formula of a compound whose empirical formula is CH2O and its molar mass is 180.

Step 1: Empirical formula mass of CH2O = 12 + 2 + 16 = 30. Step 2: n = molar mass / empirical formula mass = 180 / 30 = 6. Step 3: Molecular formula = n x empirical formula = 6 x CH2O = C6H12O6.

2. Questions for Short Answers

iii. How many molecules are present in 1.5 g H2O?

Step 1: Molar mass of H2O = 18 g/mol. Step 2: Moles = 1.5 / 18 = 0.0833 mol. Step 3: Molecules = 0.0833 x 6.022 x 10^23 = 5.02 x 10^22 molecules.

2. Questions for Short Answers

iv. What is the difference between a mole and Avogadro's number?

A mole is a unit for the amount of a substance: one mole of any substance contains the same number of particles as there are atoms in 12 g of carbon-12. Avogadro's number is that number itself, 6.022 x 10^23, and it is a pure number, not a unit. In short, the mole is the unit and Avogadro's number is the count of particles in one mole.

2. Questions for Short Answers

v. Write down the chemical equation of the following reaction. Copper + Sulphuric acid gives Copper sulphate + Sulphur dioxide + Water.

Step 1: Word equation: Cu + H2SO4 gives CuSO4 + SO2 + H2O. Step 2: Balance sulphur: two molecules of acid are needed, Cu + 2H2SO4. Step 3: Check atoms:

left Cu 1, H 4, S 2, O 8; right Cu 1, H 4, S 2, O 8. Balanced equation: Cu + 2H2SO4 gives CuSO4 + SO2 + 2H2O.

3. Constructed Response Questions

i. Different compounds will never have the same molecular formula but they can have the same empirical formula. Explain.

The molecular formula gives the exact number of atoms of each element in one molecule, and two different compounds cannot have exactly the same molecular formula and still be different compounds unless they are isomers, in which case the arrangement differs. The empirical formula gives only the simplest whole-number ratio of the atoms, so many different compounds share the same ratio. For example, ethyne C2H2 and benzene C6H6 both have the empirical formula CH, and glucose C6H12O6 and formaldehyde CH2O both have the empirical formula CH2O, although their molecules are quite different.

3. Constructed Response Questions

ii. Write down the chemical formulas of the following compounds. Calcium phosphate, Aluminium nitride, Sodium acetate, Ammonium carbonate and Bismuth sulphate.

Calcium phosphate: Ca2+ and PO4 3- combine as Ca3(PO4)2. Aluminium nitride: Al3+ and N3- combine as AlN. Sodium acetate: Na+ and CH3COO- combine as CH3COONa. Ammonium carbonate: NH4+ and CO3 2- combine as (NH4)2CO3. Bismuth sulphate: Bi3+ and SO4 2- combine as Bi2(SO4)3.

4. Descriptive Questions

i. Which conditions must be fulfilled before writing a chemical equation for a reaction?

First, the correct chemical formulas of all reactants and products must be known.

Second, the equation must be balanced, so that the number of atoms of each element is the same on both sides, obeying the law of conservation of mass. Third, the physical states of the substances should be shown, and any conditions of the reaction, such as heat, catalyst or pressure, should be written above the arrow.

4. Descriptive Questions

ii. Explain the concepts of Avogadro's number and mole.

Avogadro's number, 6.022 x 10^23, is the number of particles in one mole of any substance. It is also the number of atoms in exactly 12 g of carbon-12. The mole is the unit used to count very large numbers of atoms, molecules or ions: one mole of a substance is the amount that contains Avogadro's number of particles, and its mass in grams equals its relative atomic or molecular mass. For example, one mole of water is 18 g and contains 6.022 x 10^23 molecules.

4. Descriptive Questions

iii. How many grams of CO2 will be produced when we react 10 g of CH4 with excess of O2 according to the following equation? CH4 + 2O2 gives CO2 + 2H2O.

Step 1: Molar mass of CH4 = 16 g/mol. Step 2: Moles of CH4 = 10 / 16 = 0.625 mol. Step 3: From the equation, 1 mol of CH4 gives 1 mol of CO2, so moles of CO2 = 0.625 mol. Step 4: Mass of CO2 = 0.625 x 44 = 27.5 g.

4. Descriptive Questions

iv. How many moles of coal are needed to produce 10 moles of CO according to the following equation? 3C + O2 + H2O gives H2 + 3CO.

Step 1: From the equation, 3 mol of carbon produce 3 mol of carbon monoxide. Step 2: The ratio of C to CO is 3 to 3, that is, 1 to 1. Step 3: Moles of C needed

for 10 mol of CO = 10 x 1 / 1 = 10 mol. So 10 moles of coal (carbon) are needed.

4. Descriptive Questions

v. How much SO2 is needed in grams to produce 10 moles of sulphur? 2H2S + SO2 gives 2H2O + 3S.

Step 1: From the equation, 1 mol of SO2 produces 3 mol of sulphur. Step 2: Moles of SO2 needed for 10 mol of S = 10 / 3 = 3.33 mol. Step 3: Molar mass of SO2 = 64 g/mol. Step 4: Mass of SO2 = 3.33 x 64 = 213.3 g.

4. Descriptive Questions

vi. How much ammonia is needed in grams to produce 1 kg of urea fertilizer? 2NH3 + CO2 gives CO(NH2)2 + H2O.

Step 1: Molar mass of urea CO(NH2)2 = 60 g/mol. Step 2: Moles of urea in 1 kg = 1000 / 60 = 16.67 mol. Step 3: From the equation, 1 mol of urea needs 2 mol of NH3, so moles of NH3 = 2 x 16.67 = 33.33 mol. Step 4: Mass of NH3 = 33.33 x 17 = 566.7 g.

4. Descriptive Questions

vii. Calculate the number of atoms in the following. (a) 3 g of H2 (b) 3.4 moles of N2 (c) 10 g of C6H12O6.

(a) Moles of H2 = 3 / 2 = 1.5 mol. Molecules = 1.5 x 6.022 x 10^23 = 9.03 x 10^23. Each molecule has 2 atoms, so atoms = 2 x 9.03 x 10^23 = 1.806 x 10^24 atoms. (b) Molecules of N2 = 3.4 x 6.022 x 10^23 = 2.047 x 10^24. Each molecule has 2 atoms, so atoms = 4.095 x 10^24 atoms. (c) Moles of C6H12O6 = 10 / 180 = 0.0556 mol. Molecules = 0.0556 x 6.022 x 10^23 = 3.34 x 10^22. Each molecule has 24 atoms (6 + 12 + 6), so atoms = 24 x 3.34 x 10^22 = 8.03 x 10^23 atoms.

Chapter 5: Energetics

1. Tick the correct answer.

(i) The following reaction is an exothermic reaction. $H_2 + Cl_2$ in sunlight gives $2HCl$. From where does the energy come to break the bonds of H_2 and Cl_2?

(a) By collisions between the molecules

(b) From sunlight ✓

(c) From the surrounding

(d) By collisions of the molecules with the walls of the container

Correct answer: From sunlight

1. Tick the correct answer.

(ii) Which of the following reactions has the least value of activation energy?

(a) $H_2(g) + 1/2\ O_2(g)$ gives $H_2O(g)$

(b) $C(s) + O_2(g)$ gives $CO_2(g)$

(c) $NaCl(aq) + AgNO_3(aq)$ gives $AgCl(s) + NaNO_3(aq)$ ✓

(d) $H_2(g) + I_2(s)$ gives $2HI(g)$

Correct answer: $NaCl(aq) + AgNO_3(aq)$ gives $AgCl(s) + NaNO_3(aq)$

1. Tick the correct answer.

(iii) Formation of which hydrogen halide from the elements is an endothermic reaction?

(a) HCl

(b) HF

(c) HBr

(d) HI ✓

Correct answer: HI

1. Tick the correct answer.

(iv) Which reaction do you expect to be a reversible reaction? (Energy profiles: (a) products far below reactants; (b) products slightly above reactants; (c) products moderately above reactants; (d) products far above reactants.)

(a) Graph (a)

(b) Graph (b) ✓

(c) Graph (c)

(d) Graph (d)

Correct answer: Graph (b)

Note on Question 1 (iv)

Note: How is the reversible reaction recognised from the graphs?

A reaction is easily reversible when the energy gap between reactants and products is small, so the reverse reaction also has a small barrier to climb. In graph (b) the products lie only slightly above the reactants, so the reverse barrier is small and the reaction can readily go backwards. In graph (a) the products lie far below the reactants, so the reaction is strongly exothermic and practically goes to completion in the forward direction only.

1. Tick the correct answer.

(v) When NaOH and HCl are mixed the temperature increases. The reaction is:

(a) endothermic with a positive enthalpy change.

(b) endothermic with a negative enthalpy change.

(c) exothermic with a positive enthalpy change.

(d) exothermic with a negative enthalpy change. ✓

Correct answer: exothermic with a negative enthalpy change.

1. Tick the correct answer.

(vi) Why does the following exothermic reaction not occur? C (Diamond) gives C

(Graphite), enthalpy change = -3 kJ/mol.

(a) Structure of diamond is more stable than that of graphite.

(b) Diamond has strong covalent bonds than does the graphite.

(c) The change from diamond to graphite has high activation energy.

(d) Density of graphite is less than that of diamond.

Correct answer: The change from diamond to graphite has high activation energy.

2. Questions for Short Answers

i. What is the difference between enthalpy and enthalpy change?

Enthalpy is the total heat content of a system at constant pressure; it is a property of the substances themselves. Enthalpy change is the difference between the enthalpies of the products and the reactants, that is, the heat absorbed or given out by a reaction at constant pressure. Its symbol is delta H: negative for exothermic reactions and positive for endothermic reactions.

2. Questions for Short Answers

ii. Why is breaking of a bond an endothermic process?

Atoms in a bond are held together by attractive forces, and energy must be supplied to pull them apart against that attraction. Since energy is taken in from the surroundings to break the bond, bond breaking is always an endothermic process. Bond formation, on the other hand, releases energy and is exothermic.

2. Questions for Short Answers

iii. Depict the transition state for the following reaction. H2 + Cl2 gives 2HCl.

Draw an energy profile with energy on the vertical axis and path of reaction on the horizontal axis. The reactants H2 + Cl2 start at a low level, the curve rises to a peak, and then falls to the products 2HCl at a lower level, because the reaction

is exothermic. At the peak, draw the transition state, the activated complex, written as [H...H...Cl...Cl] with dotted lines, in which the H-H and Cl-Cl bonds are partly broken and the H-Cl bonds are partly formed. This unstable arrangement has the highest energy and is neither reactant nor product.

2. Questions for Short Answers

iv. Draw the reaction profiles for two exothermic reactions one of which moves faster than the other.

Draw two energy profiles on the same axes, both starting at the same reactant level and ending at the same lower product level, because both reactions are exothermic. The faster reaction has a lower peak, that is, a smaller activation energy, so more molecules can cross its barrier at a given temperature. The slower reaction has a higher peak, a larger activation energy, so fewer molecules can cross it and the reaction moves more slowly.

3. Constructed Response Questions

i. Explain why the reaction between atmospheric gases oxygen and nitrogen does not take place under normal conditions? But in the presence of lightning these gases react to give NO. The reaction stops as soon as the lightning stops.

Oxygen and nitrogen do not react under normal conditions because the reaction has a very high activation energy: the strong triple bond in nitrogen molecules and the double bond in oxygen molecules must first be broken, and ordinary collisions do not supply enough energy. Lightning supplies a huge amount of energy, so the molecules cross the barrier and react to form nitric oxide. As soon as the lightning stops, no more energy is supplied, the molecules can no longer cross the barrier, and the reaction stops.

3. Constructed Response Questions

ii. A reaction between natural gas (CH4) and atmospheric oxygen does not take place when you mix them. As soon as you show a burning match stick, the reaction starts immediately and then it continues until one or both of the reactants is/are used up. Explain.

Methane and oxygen do not react on simple mixing because the reaction has a high activation energy, and ordinary molecular collisions cannot break the C-H and O=O bonds. The burning match supplies the activation energy, so the reaction starts. Once started, the reaction is strongly exothermic, and the heat it releases supplies the activation energy for the next molecules, so the reaction sustains itself and continues until one or both reactants are used up.

4. Descriptive Questions

i. Explain the difference between the terms heat and enthalpy.

Heat is energy in transit: it is the energy transferred from one body to another because of a difference in temperature, and it is not stored in a body. Enthalpy is a property of a system: it is the total heat content of the system at constant pressure. When a reaction takes place at constant pressure, the heat absorbed or given out by the system equals its change in enthalpy, but heat itself is only the energy that flows during the change.

4. Descriptive Questions

ii. Explain giving examples of exothermic and endothermic reactions.

An exothermic reaction gives out heat to the surroundings, so the temperature rises and delta H is negative. Examples are the combustion of methane or petrol, the neutralisation of an acid with a base, and respiration in living cells. An endothermic reaction takes in heat from the surroundings, so the temperature falls and delta H is positive. Examples are photosynthesis in plants, the melting of ice, the thermal decomposition of calcium carbonate into lime and carbon

dioxide, and the dissolving of ammonium chloride in water.

4. Descriptive Questions

iii. How does a reaction take place?

A reaction takes place when reactant particles collide with one another. Only collisions with enough energy, at least equal to the activation energy, and with the correct orientation are effective. In an effective collision the old bonds break and new bonds form through an unstable arrangement called the transition state or activated complex, which sits at the top of the energy barrier. The activated complex then breaks down into the products, and the net energy difference between reactants and products appears as heat absorbed or given out.

4. Descriptive Questions

iv. Explain the following terms: Activation energy, Transition state, system and surrounding.

Activation energy is the minimum extra energy that reactant molecules must gain before they can react; it is the height of the energy barrier between reactants and products. The transition state is the unstable, highest-energy arrangement of atoms at the top of that barrier, in which old bonds are partly broken and new bonds are partly formed. The system is the part of the universe under study, for example the reacting mixture in a beaker. The surroundings are everything else around the system that can exchange energy with it.

Chapter 6: Equilibria

1. Tick the correct answer.

(i) What will happen if the rates of forward and reverse reactions are very high?

(a) The equilibrium point will reach very soon. ✓

(b) The equilibrium point will reach very late.

(c) The reaction will not attain the state of dynamic equilibrium.

(d) The reaction will be practically irreversible.

Correct answer: The equilibrium point will reach very soon.

1. Tick the correct answer.

(ii) Predict which components of the atmosphere react in the presence of lightning.

(a) N_2 and H_2O

(b) O_2 and H_2O

(c) CO_2 and O_2

(d) N_2 and O_2 ✓

Correct answer: N_2 and O_2

1. Tick the correct answer.

(iii) CaO or lime is used extensively in steel, glass and paper industries. It is produced in a reversible reaction by the decomposition of lime stone ($CaCO_3$). Choose the conditions to produce maximum amount of lime.

(a) Heating at high temperature in a closed vessel

(b) Heating at high temperature in an open vessel ✓

(c) Cooling it in a closed vessel

(d) Cooling it in an open vessel

Correct answer: Heating at high temperature in an open vessel

Note on Question 1 (iii)

Note: The question calls the decomposition of limestone an exothermic reaction, which is an error in the book.

The decomposition of calcium carbonate, $CaCO_3$ gives $CaO + CO_2$, is in fact endothermic: heat must be supplied to break limestone down. The correct conditions for maximum lime are therefore a high temperature, which favours the endothermic forward reaction, and an open vessel, so that carbon dioxide escapes and the equilibrium is pulled forward. Option (b) is the correct choice on this correct reasoning, even though the question wrongly labels the reaction exothermic.

1. Tick the correct answer.

(iv) What condition should be met for the reversible reaction to achieve the state of equilibrium?

(a) All the reactants should be converted into the products.

(b) 50% of the reactants should be converted into products.

(c) The concentrations of all the reactants and the products should become constant. ✓

(d) One of the products should be removed from the reaction mixture.

Correct answer: The concentrations of all the reactants and the products should become constant.

1. Tick the correct answer.

(v) Why the gas starts coming out when you open a can of fizzy drink?

(a) Because the solubility of the gas increases

(b) Because the gas is insoluble in water

(c) Because the gas is dissolved under pressure hence it comes out when pressure is decreased ✓

(d) Because the solubility of the gas decreases at high pressure

Correct answer: Because the gas is dissolved under pressure hence it comes out when pressure is decreased

1. Tick the correct answer.

(vi) The following reaction is performed in a closed vessel. $CaCO_3(s)$ gives $CaO(s) + CO_2(g)$. How the equilibrium will be affected if you increase the pressure?

(a) The forward reaction will be favoured

(b) The backward reaction will be favoured ✓

(c) No effect on backward reaction

(d) No effect on forward or backward reaction

Correct answer: The backward reaction will be favoured

1. Tick the correct answer.

(vii) Is reversible reaction useful for preparing compounds on large scale?

(a) No

(b) Yes

(c) They are useful only when equilibrium lies far to the right side ✓

(d) They are useful only when equilibrium lies far to the left side

Correct answer: They are useful only when equilibrium lies far to the right side

2. Questions for Short Answers

i. How is dynamic equilibrium different from the static equilibrium?

In dynamic equilibrium the forward and reverse reactions both continue to take place, but at equal rates, so the concentrations of reactants and products stay constant although the reactions never stop. In static equilibrium nothing at all happens: no reaction takes place in either direction, as in a book resting on a

table, where the forces balance but there is no movement or change.

2. Questions for Short Answers

ii. How the following reversible reaction will be affected if its temperature is increased? 2H2O(l) with electricity gives 2H2(g) + O2(g).

The decomposition of water into hydrogen and oxygen is an endothermic reaction, because energy must be supplied to split water. Increasing the temperature therefore favours the forward reaction, and more water decomposes to give more hydrogen and oxygen, shifting the equilibrium to the right.

2. Questions for Short Answers

iii. How can you get the maximum yield in a reversible reaction?

Maximum yield is obtained by shifting the equilibrium as far as possible towards the products, following Le Chatelier's principle: add more reactants, remove a product as soon as it forms, choose the temperature that favours the forward reaction, and, where the number of gas molecules changes, choose the pressure that favours the side with fewer gas molecules. Using a catalyst does not change the yield; it only helps equilibrium to be reached faster.

2. Questions for Short Answers

iv. How can you decrease the time to attain the position of equilibrium in a reversible reaction?

Equilibrium is reached faster by increasing the rates of both the forward and the reverse reactions. This is done by adding a catalyst, by raising the temperature, and by increasing the concentrations of the reactants or the pressure of the gases. The position of equilibrium itself is not changed by a catalyst; only the time needed to reach it becomes shorter.

2. Questions for Short Answers

v. What is the effect of increasing pressure on the following reaction at equilibrium? N2(g) + O2(g) gives 2NO(g).

There is no effect. The left side has two moles of gas (one of nitrogen plus one of oxygen) and the right side also has two moles of gas (two of nitric oxide). Since the number of gas molecules is the same on both sides, changing the pressure does not shift the equilibrium in either direction.

3. Constructed Response Questions

i. Why are some reactions irreversible while others are reversible?

A reaction is irreversible when its products cannot easily react back to give the reactants, usually because a product escapes from the mixture as a gas, separates as an insoluble precipitate, or because the reaction releases so much energy that the reverse barrier is too high. A reaction is reversible when it takes place in a closed system, its products remain in contact with each other, and the energy barrier for the reverse reaction is small enough for the products to react back.

3. Constructed Response Questions

ii. Why are combustion reactions generally irreversible?

Combustion reactions are generally irreversible because their products, carbon dioxide and water vapour, escape into the air together with a large amount of heat and light. The products are no longer held together in a closed system, and they cannot recombine on their own to form the fuel again. The reaction therefore goes practically to completion in the forward direction.

3. Constructed Response Questions

iii. How do you know if a reaction is reversible or irreversible?

A reaction is reversible if, in a closed vessel, it does not go to completion but reaches a state where reactants and products are both present in constant amounts, and if changing the conditions, such as temperature, pressure or concentration, shifts the mixture back or forward. A reaction is irreversible if it goes to completion, if a product escapes or precipitates out, and if the original reactants cannot be recovered from the products by changing the conditions.

4. Descriptive Questions

i. How can you drive the reversible reaction at equilibrium? (a) in the forward direction (b) in the backward direction.

(a) To drive the reaction forward: add more reactant, remove a product as it forms, raise the temperature if the forward reaction is endothermic (or lower it if the forward reaction is exothermic), and raise the pressure if the forward side has fewer gas molecules. (b) To drive it backward: do the opposite, that is, add more product, remove a reactant, change the temperature in the opposite direction, or change the pressure to favour the side with more gas molecules.

4. Descriptive Questions

ii. Explain how the rates of forward and backward reactions change when the system approaches equilibrium.

At the start only reactants are present, so the forward reaction is fast and the reverse reaction cannot take place at all. As reactants are used up and products build up, the forward rate falls because the reactant concentration falls, while the reverse rate rises because the product concentration rises. The two rates keep approaching each other until they become equal. At that point equilibrium is reached, and both reactions continue at the same rate, so no further change in concentration is seen.

4. Descriptive Questions

iii. Describe the effect of a catalyst on the reversible reaction.

A catalyst speeds up both the forward and the reverse reactions by the same factor, because it lowers the activation energy of both directions equally. Equilibrium is therefore reached much faster, but the position of equilibrium and the final amounts of reactants and products remain exactly the same as they would be without the catalyst.

4. Descriptive Questions

iv. How can a reversible reaction be forced to go to completion?

A reversible reaction can be forced to completion by continuously removing one of its products from the mixture, so that the reverse reaction has nothing to react back with and the forward reaction keeps going. This is done by letting a gaseous product escape, by precipitating a product as an insoluble solid, or by using a large excess of one reactant. In each case Le Chatelier's principle keeps pulling the equilibrium forward until the other reactant is used up.

4. Descriptive Questions

v. How does change in temperature affect the reaction at equilibrium?

Increasing the temperature favours the endothermic direction of a reversible reaction, and decreasing the temperature favours the exothermic direction. For example, in an exothermic forward reaction, heating shifts the equilibrium backwards towards the reactants, while cooling shifts it forwards towards the products. Temperature is the only factor that changes the value of the equilibrium constant itself; concentration, pressure and catalysts only shift the position or the speed.

Chapter 7: Acid Base Chemistry

1. Tick the correct answer.

(i) Which acid is not used as a food or mixed with food?

(a) Tartaric acid

(b) Ascorbic acid

(c) Citric acid

(d) Formic acid ✓

Correct answer: Formic acid

1. Tick the correct answer.

(ii) While baking, which gas is responsible for raising the bread and making it soft?

(a) Oxygen

(b) Carbon dioxide ✓

(c) Nitrogen

(d) Carbon monoxide

Correct answer: Carbon dioxide

1. Tick the correct answer.

(iii) How many hydroxide ions, calcium hydroxide will release in water?

(a) 1

(b) 2 ✓

(c) Zero

(d) 3

Correct answer: 2

1. Tick the correct answer.

(iv) In a neutralization reaction between KOH and H3PO4, how many molecules

of KOH will react with one molecule of H3PO4?

(a) 2

(b) 1

(c) 3 ✓

(d) 4

Correct answer: 3

1. Tick the correct answer.

(v) Which compound is formed when SO2 is dissolved in water?

(a) SO3

(b) H2SO3 ✓

(c) H2SO4

(d) H2S2O7

Correct answer: H2SO3

1. Tick the correct answer.

(vi) Which compound in the following reaction is behaving as a conjugate base?

CH3COOH(aq) + H2O(l) gives CH3COO-(aq) + H3O+(aq).

(a) CH3COOH

(b) H2O

(c) CH3COO- ✓

(d) H3O+

Correct answer: CH3COO-

2. Questions for Short Answers

i. Choose Arrhenius Acids among the following compounds. HF, NH4+, H2SO3, SO3, H2S, H2O.

The Arrhenius acids are HF, H2SO3 and H2S, because each of these releases

hydrogen ions (H+) when dissolved in water. NH4+ is not an Arrhenius acid; it donates a proton only in the Bronsted-Lowry sense. SO3 is an acidic oxide, not an acid itself, although it forms sulphuric acid with water. H2O is neutral and amphoteric, acting as an acid or a base as the situation requires.

2. Questions for Short Answers

ii. How does calcium metal react with dilute H2SO4?

Calcium reacts with dilute sulphuric acid to form calcium sulphate and hydrogen gas: Ca + H2SO4 gives CaSO4 + H2. The reaction starts vigorously but soon slows down, because the calcium sulphate formed is only sparingly soluble and coats the metal, partly stopping further attack by the acid.

2. Questions for Short Answers

iii. Which salt is formed when HCl reacts with BaCO3?

Barium chloride is formed, together with water and carbon dioxide: BaCO3 + 2HCl gives BaCl2 + H2O + CO2. The carbonate is decomposed by the acid, and the barium salt remains in solution.

2. Questions for Short Answers

iv. How will you justify that HSO4- is a Bronsted-Lowry acid?

A Bronsted-Lowry acid is a substance that donates a proton (H+) to another substance. The hydrogen sulphate ion donates a proton to water or to a base, forming the sulphate ion: HSO4- gives SO4 2- + H+. Because it donates a proton in this way, HSO4- behaves as a Bronsted-Lowry acid, although it can also accept a proton and act as a base.

2. Questions for Short Answers

v. Why is HCl not edible although it is present in the stomach and responsible for

digestion of food?

The hydrochloric acid in the stomach is very dilute, about 0.1 M, and the stomach wall is protected by a thick layer of mucus that stops the acid from attacking the tissue. Drinking hydrochloric acid would mean taking a concentrated, corrosive acid that would burn the mouth, throat and stomach lining, because those parts have no such protection. The acid is safe only in its dilute form and in its proper place.

3. Constructed Response Questions

i. What chemical name will you give to soap as a compound?

Soap is the sodium or potassium salt of a long-chain fatty acid. Common soap is sodium stearate, the sodium salt of stearic acid, with the formula C17H35COONa. It is formed when a fat or oil is boiled with sodium hydroxide, a process called saponification.

3. Constructed Response Questions

ii. In the presence of a drop of an acid, water is known to ionize as follows: H2O gives H+ + OH-. In your opinion, which name will be suitable for water: an acid, a base or both?

Both. Water is amphoteric: it can act as an acid by donating a proton to form the hydroxide ion, and it can act as a base by accepting a proton to form the hydroxonium ion. In pure water one molecule donates a proton to another, giving H3O+ and OH- in equal numbers, which is why pure water is neutral.

3. Constructed Response Questions

iii. Why does Na2CO3 behave like a base in water?

Sodium carbonate dissolves to give sodium ions and carbonate ions. The carbonate ion reacts with water, taking a proton from it: CO3 2- + H2O gives

HCO3- + OH-. The hydroxide ions produced make the solution basic, so sodium carbonate behaves like a base in water even though it contains no hydroxide group itself. Salts of this kind are said to undergo hydrolysis.

4. Descriptive Questions

i. Explain Arrhenius concept of acids and bases.

According to Arrhenius, an acid is a substance that releases hydrogen ions (H+) when dissolved in water, for example HCl gives H+ + Cl-, and H2SO4 gives 2H+ + SO4 2-. A base is a substance that releases hydroxide ions (OH-) in water, for example NaOH gives Na+ + OH-. An acid and a base neutralise each other to form a salt and water, because the H+ and OH- ions combine to form water. The concept is limited because it applies only to water solutions and cannot explain substances like ammonia, which acts as a base without containing OH-.

4. Descriptive Questions

ii. Compare Arrhenius and Bronsted-Lowry concepts of acids and bases.

Arrhenius defines an acid as a producer of H+ ions in water and a base as a producer of OH- ions in water, so it works only in water solution. Bronsted-Lowry defines an acid as a proton donor and a base as a proton acceptor, in any solvent and even without water. Every Arrhenius acid is also a Bronsted-Lowry acid, but the Bronsted-Lowry concept is wider: it covers ammonia as a base, since NH3 accepts a proton to become NH4+, and it explains conjugate acid-base pairs, such as CH3COOH and CH3COO-.

4. Descriptive Questions

iii. How does sulphuric acid react with the following compounds? NH4Cl, NH4NO3, MgO, MgCO3.

With ammonium chloride: H2SO4 + NH4Cl gives NH4HSO4 + HCl, and hydrogen

chloride gas is released. With ammonium nitrate: H2SO4 + NH4NO3 gives NH4HSO4 + HNO3, and nitric acid is released. With magnesium oxide: H2SO4 + MgO gives MgSO4 + H2O, a neutralisation reaction giving magnesium sulphate. With magnesium carbonate: H2SO4 + MgCO3 gives MgSO4 + H2O + CO2, giving magnesium sulphate with the release of carbon dioxide gas.

4. Descriptive Questions

iv. What happens when a base reacts with a non-metallic oxide. What do you infer about the nature of non-metallic oxide?

When a base reacts with a non-metallic oxide, a salt and water are formed, just as in neutralisation. For example, 2NaOH + CO2 gives Na2CO3 + H2O, and Ca(OH)2 + SO2 gives CaSO3 + H2O. Since the non-metallic oxide behaves like an acid towards a base, we infer that non-metallic oxides are acidic in nature. They are therefore called acidic oxides, and they form acids when dissolved in water.

4. Descriptive Questions

v. State the reason of showing acidic character by both dry HCl gas and HCl solution in water.

Dry hydrogen chloride gas shows no acidic character by itself, because acidity needs free hydrogen ions, and dry HCl molecules do not release H+ ions. It does not turn dry blue litmus paper red. When HCl is dissolved in water, its molecules ionise completely to give H+ (as H3O+) and Cl- ions, and these hydrogen ions give the solution its acidic character: it turns blue litmus red and reacts with bases and metals. In short, the gas is not acidic, but its water solution is strongly acidic.

4. Descriptive Questions

vi. Differentiate between an acid and its conjugate base.

An acid and its conjugate base differ by exactly one proton. The acid is the species that donates the proton, and the conjugate base is what remains after the proton has been donated. For example, in the pair CH3COOH / CH3COO-, acetic acid is the acid and the acetate ion is its conjugate base. A strong acid has a weak conjugate base, and a weak acid has a comparatively strong conjugate base, because the two members of the pair are linked by the gain or loss of one H+ ion.

Chapter 8: Periodic Table and Periodicity

In-chapter Exercise (book p. 85)

(1) Barium (Ba) is present in 2nd group and 6th period. Answer the following questions about this element. (i) Is it a metal or a non-metal? (ii) Will it be electropositive or electronegative? (iii) What is the nature of its oxide? (iv) In which physical state you expect this element to exist?

(i) Barium is a metal; it lies in group 2, the alkaline earth metals. (ii) It is electropositive, because like other group 2 metals it loses its two outer electrons readily to form Ba2+ ions. (iii) Its oxide, BaO, is basic in nature, as is usual for metal oxides. (iv) It is a solid at room temperature, a soft silvery-white metal.

In-chapter Exercise (book p. 85)

(2) In which group and period you expect to find an element with largest atomic radius?

At the bottom left of the periodic table: group 1 and period 7. Atomic radius increases down a group, because new shells are added, and decreases across a period, because the nuclear charge pulls the electrons in. The largest atom is therefore the lowest element of the first group, francium.

In-chapter Exercise (book p. 85)

(3) Can you predict the group number of the most electropositive and the most electronegative elements?

Yes. The most electropositive elements are the alkali metals of group 1, which lose their single outer electron most easily; francium and caesium are the most electropositive. The most electronegative element is fluorine, which lies in group 17, the halogens, which need only one electron to complete their outer shell.

In-chapter Exercise (book p. 85)

(4) Choose among the following the element having the lowest ionization energy and the element with highest electron affinity. Also assign its group number and period number. Li, K, O, F, Cl.

Lowest ionisation energy: potassium (K), group 1, period 4, because it is the lowest alkali metal in the list and loses its outer electron most easily. Highest electron affinity: chlorine (Cl), group 17, period 3. Chlorine gains an electron more readily than fluorine in energy terms, because in tiny fluorine atoms the added electron is crowded by the electrons already present, while chlorine has room for it.

In-chapter Exercise (book p. 85)

(5) Which two elements of the periodic table react to give (i) A basic oxide and (ii) An acidic oxide?

(i) A basic oxide is formed when a metal reacts with oxygen, for example sodium burns in oxygen to give sodium oxide, Na2O, which is basic. (ii) An acidic oxide is formed when a non-metal reacts with oxygen, for example sulphur burns in oxygen to give sulphur dioxide, SO2, which is acidic.

1. Tick the correct answer.

(i) Identify the electronic configuration of the outermost shell of a transition metal.

(a) ns2 np4

(b) ndx ns2 

(c) ns2 np6

(d) ns2 np5

Correct answer: ndx ns2

1. Tick the correct answer.

(ii) Which is the softest metal?

(a) Na ✓

(b) Ca

(c) Al

(d) Zn

Correct answer: Na

1. Tick the correct answer.

(iii) A yellow solid element exists in allotropic forms which is also present in fossil fuels. Indicate the name.

(a) Carbon

(b) Iodine

(c) Aluminium

(d) Sulphur ✓

Correct answer: Sulphur

1. Tick the correct answer.

(iv) How many electrons can nitrogen accept in its outermost shell?

(a) 2

(b) 3 ✓

(c) 4

(d) 5

Correct answer: 3

1. Tick the correct answer.

(v) Which element is the most reactive element?

(a) Oxygen

(b) Chlorine

(c) Fluorine ✓

(d) Nitrogen

Correct answer: Fluorine

1. Tick the correct answer.

(vi) Which element has the highest melting point?

(a) Na ✓

(b) K

(c) Rb

(d) Cs

Correct answer: Na

1. Tick the correct answer.

(vii) Which of the following best describe the correct order of oxygen, fluorine, and nitrogen's atomic radii?

(a) O < F < N

(b) N < F < O

(c) F < O < N ✓

(d) O < N < F

Correct answer: F < O < N

2. Questions for Short Answers

i. Why was atomic number chosen to arrange the elements in the periodic table?

Atomic number was chosen because the properties of elements repeat periodically with atomic number, as Moseley showed. Arranging by atomic mass caused problems: argon (mass 40) would come after potassium (mass 39) although argon belongs before it, and cobalt and nickel, tellurium and iodine showed similar inversions. Atomic number removed all these anomalies and

gave every element its correct place.

2. Questions for Short Answers

ii. What is the significance of the word periodic?

Periodic means repeating at regular intervals. In the periodic table, elements with similar properties recur at regular intervals when they are arranged in order of atomic number, because their outer electronic configurations repeat. This repetition of properties is called periodicity.

2. Questions for Short Answers

iii. Why does the size of a period increase as we move down the periodic table?

Each period fills a new set of sub-shells, and the higher periods have more sub-shells available to fill. The first period fills only the 1s sub-shell and holds 2 elements, the second and third periods fill s and p sub-shells and hold 8 each, the fourth and fifth also fill d sub-shells and hold 18 each, and the sixth also fills f sub-shells and holds 32. Periods therefore grow longer as we move down the table.

2. Questions for Short Answers

iv. In a group, the elements have the same number of electrons in the outermost shell. Why is it so?

Elements are placed in a group according to their outer electronic configuration. When atomic number increases down a group, each new element adds a whole new shell, but the number of electrons in the outermost shell stays the same, because the group number itself is fixed by that outer-shell count. Hence all elements of one group have the same number of valence electrons and similar chemical properties.

2. Questions for Short Answers

v. Which element has the maximum atomic radius and which element has the minimum atomic radius in third period?

In the third period, sodium (Na) has the maximum atomic radius and chlorine (Cl) has the minimum atomic radius, argon being a noble gas left out of the comparison. Across a period the nuclear charge increases and pulls the electron shells inward, so the radius decreases from left to right.

2. Questions for Short Answers

vi. The first ionization energy value of magnesium is less than the second one. Give reason.

The first ionisation energy removes an electron from a neutral magnesium atom, Mg, to give Mg+. The second ionisation energy removes an electron from Mg+, which already carries a positive charge and holds its remaining electrons more tightly, and which has the stable configuration 2,8. More energy is therefore needed to remove the second electron than the first.

2. Questions for Short Answers

vii. Which element has the least value of ionization energy and which element has the highest value of electronegativity?

The least ionisation energy belongs to the alkali metals at the bottom left of the table, caesium (and francium), because their single outer electron is far from the nucleus and well shielded, so it is lost very easily. The highest electronegativity belongs to fluorine, at the top right of the table, because its small atom attracts a shared pair of electrons more strongly than any other element.

3. Constructed Response Questions

i. Suppose a new element is discovered. Where would you like to accommodate

this element in the periodic table?

The new element would be placed by its atomic number, in the position following the element before it, because atomic number fixes the order of the table. Its group would be fixed by the number of electrons in its outermost shell, and its period by the number of shells its electrons occupy. If its electronic configuration were known, its exact block and neighbours would follow at once.

3. Constructed Response Questions

ii. What is the first element of the periodic table? Will it lose an electron or gain it?

The first element of the periodic table is hydrogen, with atomic number 1 and one electron in its first shell. Hydrogen can do either: it can lose its electron to form the positive hydrogen ion H+, as in acids, or it can gain one electron to complete its shell and form the negative hydride ion H-, as in sodium hydride. Its behaviour depends on the element it reacts with.

3. Constructed Response Questions

iii. Why is it not possible for oxygen atom to accept three electrons to form O3- ion like nitrogen which can accept electrons to form N3-?

Nitrogen has five electrons in its outer shell, so accepting three more completes its octet of eight, giving N3-. Oxygen already has six outer electrons and needs only two more to complete its octet, giving O2-. Accepting a third electron would give oxygen ten electrons around it, more than an octet, and the second shell has no d orbitals to hold extra electrons. Moreover, adding a third negative charge to an already doubly negative ion is opposed by strong repulsion. O3- therefore cannot form.

4. Descriptive Questions

i. Which information is needed to locate the elements in the periodic table if you do not know its atomic number? Is atomic mass helpful for this purpose?

To locate an element without its atomic number, we need its electronic configuration, especially the number of shells and the number of outer-shell electrons, together with its chemical properties, such as its valency and the kind of compounds it forms. Atomic mass is not reliable for this purpose: isotopes of one element have different masses, and the mass order itself misplaces some elements, as with argon and potassium and with cobalt and nickel.

4. Descriptive Questions

ii. How many blocks of elements are present in the periodic table? Are these blocks helpful in studying the properties of elements?

There are four blocks in the periodic table: the s-block, the p-block, the d-block and the f-block, named after the sub-shell into which the last electron enters. Yes, the blocks are very helpful: the s-block holds reactive metals, the p-block holds most non-metals and metalloids, the d-block holds the transition metals, and the f-block holds the inner transition metals. Knowing the block at once suggests the valency, metallic character and typical compounds of an element.

4. Descriptive Questions

iii. Explain the variation in the following properties in the periods giving reasons. (a) Atomic radius (b) Ionization energy.

(a) Atomic radius decreases across a period from left to right. The electrons are added to the same shell, while the nuclear charge increases, so the nucleus pulls the whole shell inward more strongly. (b) Ionisation energy increases across a period. The same increasing nuclear pull holds the outer electrons more tightly, and the decreasing size brings them closer to the nucleus, so more energy is needed to remove an electron.

4. Descriptive Questions

iv. Which physical properties of elements may lead us to know what type of bond it will form?

Melting and boiling points, electrical conductivity, hardness, density and solubility give strong clues. A high melting point with conduction when molten points to ionic bonding; a low melting point with no conduction points to covalent molecules; a high melting point with conduction in the solid state and the ability to be beaten into sheets points to metallic bonding. Lustre and malleability further confirm the metallic character.

4. Descriptive Questions

v. Write down the names of four non-metals which exist in solid state at normal temperature.

Four non-metals that are solids at normal temperature are carbon, sulphur, phosphorus and iodine. Others are boron, silicon and selenium. Most non-metals are gases at room temperature, but these have strong bonds or strong forces between their particles that keep them solid.

Chapter 9: Group Properties and Elements

1. Tick the correct answer.

(i) Which halogen will have the least reactivity with alkaline earth metals?

(a) Chlorine

(b) Iodine ✓

(c) Bromine

(d) Fluorine

Correct answer: Iodine

1. Tick the correct answer.

(ii) Which compound do you expect to be coloured?

(a) KCl

(b) $BaCl_2$

(c) $AlCl_3$

(d) $NiCl_2$ ✓

Correct answer: $NiCl_2$

1. Tick the correct answer.

(iii) In which element there exists the strongest forces of attraction between atoms?

(a) Mg ✓

(b) Ca

(c) Sr

(d) Ba

Correct answer: Mg

1. Tick the correct answer.

(iv) Which halogen acid is unstable at room temperature?

(a) HBr

(b) HI ✓

(c) HCl

(d) HF

Correct answer: HI

1. Tick the correct answer.

(v) Which oxide is the most basic oxide?

(a) Na2O ✓

(b) Li2O

(c) MgO

(d) CO

Correct answer: Na2O

1. Tick the correct answer.

(vi) The following solutions of a halogen and a sodium halide are mixed together.

Which solution will turn dark because of a reaction?

(a) Br2 and NaCl

(b) Br2 and NaF

(c) Cl2 and NaF

(d) Cl2 and NaI ✓

Correct answer: Cl2 and NaI

1. Tick the correct answer.

(vii) X is a monoatomic gas, which statement about this is correct?

(a) X burns in air

(b) X is coloured

(c) X is unreactive ✓

(d) X will displace iodine from it

Correct answer: X is unreactive

2. Questions for Short Answers

i. Why does it become easier to cut an alkali metal when we move from top to bottom in a group I?

As we move down group 1 the atoms become larger, because a new shell is added at each step, and the metallic bond between them becomes weaker. The metal therefore becomes softer and easier to cut: lithium is fairly hard, sodium cuts easily, and potassium and caesium are softer still.

2. Questions for Short Answers

ii. Predict the reactivity of potassium towards halogens.

Potassium will react very vigorously with halogens, even more violently than sodium. It loses its single outer electron very easily, and the halogen gains it readily, so the reaction is fast and strongly exothermic. With chlorine, for example, potassium burns with a lilac flame to form potassium chloride, KCl.

2. Questions for Short Answers

iii. In the following reaction, chlorine acts as an oxidising agent. Which is the reducing agent? Cl2(aq) + 2NaBr(aq) gives 2NaCl(aq) + Br2(g).

The reducing agent is sodium bromide, that is, the bromide ion Br-. Chlorine is reduced as it gains electrons to become chloride ions, while the bromide ions lose electrons and are oxidised to bromine. The substance that is itself oxidised is the reducing agent, so NaBr is the reducing agent.

2. Questions for Short Answers

iv. Why does iodine exist in the solid state at room temperature?

Iodine molecules are large and contain many electrons, so the London dispersion forces between them are strong. A large amount of heat energy is needed to separate the molecules, so iodine remains a solid at room temperature, whereas the smaller halogens chlorine and fluorine, with much weaker forces, are gases.

3. Constructed Response Questions

i. Which noble gas should have the lowest boiling point and why?

Helium should have the lowest boiling point of all the noble gases. Helium atoms are the smallest and contain only two electrons, so the forces between them are the weakest of all. Almost no energy is needed to separate helium atoms, so helium boils at -269 C, lower than any other element.

3. Constructed Response Questions

ii. Why are almost all the metals solids while non-metals generally exist as gases and solids?

Metals are solids because the strong metallic bond holds their atoms firmly in a regular lattice, and a large amount of energy is needed to break that lattice. Non-metals form small covalent molecules held together only by weak intermolecular forces, so many of them are gases at room temperature. Those non-metals that form giant covalent structures, like carbon and sulphur, or large molecules, like iodine, are solids.

3. Constructed Response Questions

iii. Compare the reactivity of chlorine and bromine as an oxidising agent.

Chlorine is a stronger oxidising agent than bromine. Chlorine atoms are smaller and more electronegative, so they attract and gain electrons more readily than bromine atoms. This is shown directly by the displacement reaction: chlorine displaces bromine from sodium bromide, giving sodium chloride and bromine,

while bromine cannot displace chlorine from sodium chloride.

4. Descriptive Questions

i. Explain the role of catalytic converter in an automobile.

A catalytic converter is fitted in the exhaust system of a car to clean up the exhaust gases before they reach the air. Its catalyst, usually platinum, palladium and rhodium, converts poisonous carbon monoxide into harmless carbon dioxide, converts unburnt hydrocarbons into carbon dioxide and water, and converts harmful nitrogen oxides into harmless nitrogen gas. The catalyst itself is not used up, and the converter greatly reduces air pollution from vehicles.

4. Descriptive Questions

ii. Why do the chemical reactivities of alkali metals increase down the group whereas they decrease down the group in case of halogens?

Alkali metals react by losing their outer electron. Down the group the atoms get larger, the outer electron is farther from the nucleus and better shielded, so it is lost more easily and reactivity increases. Halogens react by gaining an electron. Down the group the atoms get larger, the incoming electron is farther from the nucleus and more shielded, so it is gained less easily and reactivity decreases. The two groups therefore show opposite trends.

4. Descriptive Questions

iii. Why are metals generally tough and strong whereas non-metals are neither tough nor strong?

Metals are tough and strong because the metallic bond holds their ions in a regular lattice with a sea of mobile electrons: when a force is applied, layers slip but the bond holds, so the metal bends instead of breaking. Non-metals have no such bond: their molecules are held only by weak intermolecular forces, or their

crystals are brittle, so they break or crumble under a force instead of bending. They are therefore neither tough nor strong.

4. Descriptive Questions

iv. Both alkali metals and halogens are very reactive elements with roles opposite to each other. Explain.

Alkali metals and halogens are both very reactive, but in opposite ways. Alkali metals have one electron in their outer shell, which they lose very easily, so they act as strong reducing agents and form positive ions. Halogens have seven electrons in their outer shell and need only one more to complete it, so they gain electrons very easily, acting as strong oxidising agents and forming negative ions. One group gives electrons, the other takes them, and together they form stable ionic compounds such as sodium chloride.

4. Descriptive Questions

v. Compare the properties of metals and non-metals.

Metals are usually lustrous solids with high melting and boiling points, except mercury, which is a liquid. They are good conductors of heat and electricity, and they can be beaten into sheets and drawn into wires. Non-metals may be solids, liquids or gases at room temperature and show a wide range of melting and boiling points. They are poor conductors of heat and electricity, except graphite, they are brittle, and they are dull and cannot be polished, except iodine. Metals are generally tough and strong; non-metals are neither tough nor strong.

Chapter 10: Environmental Chemistry

1. Tick the correct answer.

(i) Which gases are responsible for greenhouse effect?

(a) SO_2, NO_2

(b) NO_2, CO

(c) CO_2, CH_4 ✓

(d) O_2, N_2

Correct answer: CO_2, CH_4

1. Tick the correct answer.

(ii) Concentration of which gas in the atmosphere is decreased by photosynthesis in plants?

(a) Oxygen

(b) Nitrogen

(c) Carbon dioxide ✓

(d) Water vapours

Correct answer: Carbon dioxide

1. Tick the correct answer.

(iii) Which substance remains unaffected in the catalytic converter fixed in the exhaust of the automobiles?

(a) CO_2 ✓

(b) CO

(c) NO

(d) NO_2

Correct answer: CO_2

1. Tick the correct answer.

(iv) People of which age groups are most affected by the air pollution?

(a) Young adults

(b) Middle age people

(c) Children

(d) Both children and aged people ✓

Correct answer: Both children and aged people

1. Tick the correct answer.

(v) In which area there is a greater possibility of acid rain?

(a) Around village

(b) Around big cities

(c) Around industrial area ✓

(d) Around water bodies

Correct answer: Around industrial area

1. Tick the correct answer.

(vi) Which catalyst is used in the catalytic converter fixed in the exhaust systems of automobiles?

(a) Ni

(b) Cu

(c) Pt, Pd and Rh ✓

(d) CaO

Correct answer: Pt, Pd and Rh

1. Tick the correct answer.

(vii) Which gases contribute towards the formation of acid rain?

(a) Oxides of carbon

(b) Oxides of sulphur

(c) Oxides of nitrogen

(d) Both the oxides of nitrogen and sulfur

Correct answer: Both the oxides of nitrogen and sulfur

2. Questions for Short Answers

i. What is the main objective of environmental education?

The main objective of environmental education is to make people aware of the environment and its problems, and to give them the knowledge, attitudes and skills needed to protect it. An environmentally educated person understands how human actions affect air, water, soil and living things, and takes part in solving environmental problems.

2. Questions for Short Answers

ii. How is particulate matter released in the atmosphere?

Particulate matter is released by the burning of fuels in vehicles, factories and power stations, by industrial processes such as crushing and grinding, by construction work, by the burning of crop residue and wood, and by natural sources such as dust storms, forest fires and volcanic eruptions. These tiny solid and liquid particles stay suspended in the air and are breathed into the lungs.

2. Questions for Short Answers

iii. Which gas is more poisonous, CO2 or CO?

Carbon monoxide, CO, is far more poisonous than carbon dioxide. Carbon monoxide combines with the haemoglobin in red blood cells about 200 to 300 times more strongly than oxygen does, forming carboxyhaemoglobin, so the blood can no longer carry enough oxygen to the body, and a person can suffocate. Carbon dioxide is harmful mainly in large amounts, because it displaces oxygen and causes suffocation, but it does not poison the blood in this

way.

2. Questions for Short Answers

iv. How does acid rain affect forests?

Acid rain damages the leaves of trees directly, so they cannot photosynthesise properly. It also washes important nutrients such as calcium and magnesium out of the soil, and it releases harmful aluminium from the soil, which damages the roots. Trees therefore grow weakly, become open to disease, and large areas of forest can die.

2. Questions for Short Answers

v. In what way sulphur present in fossil fuels becomes dangerous?

When fossil fuels containing sulphur are burnt, the sulphur burns to form sulphur dioxide gas. Sulphur dioxide irritates the eyes, nose and lungs and causes breathing diseases, and in the air it is oxidised and dissolved in rain water to form sulphuric acid, which falls as acid rain. Acid rain damages crops, forests, buildings and water life.

2. Questions for Short Answers

vi. How is wind energy useful for us?

Wind energy is a clean and renewable source of energy that does not burn fuel and does not pollute the air. Windmills and wind turbines turn the energy of moving air into electricity for homes and industry, into mechanical work for pumping water and grinding grain, and into power for remote areas. It costs no fuel once built and helps to reduce the use of fossil fuels.

3. Constructed Response Questions

i. How is the excessive use of insecticides and pesticides harmful for birds?

Insecticides and pesticides sprayed on crops enter the food chain: insects eat the poison, and birds eat the poisoned insects, so the poison builds up in the birds' bodies. Some pesticides, such as DDT, thin the shells of birds' eggs, so the eggs break before the chicks hatch. Pesticides also kill the insects that many birds feed on, so birds starve, and the poisoned birds themselves may die.

3. Constructed Response Questions

ii. Percentage of CO2 in air is only 0.04%. Then how does it become harmful for the ecosystem?

Even at only 0.04 percent, carbon dioxide is the most important greenhouse gas produced by human activity, because it absorbs the infrared radiation given off by the Earth and traps heat in the atmosphere. It also stays in the air for hundreds of years, so every extra bit adds to the total. A small rise in its percentage therefore warms the whole planet, disturbs rainfall, melts ice and threatens the balance of the ecosystem.

3. Constructed Response Questions

iii. Why only some pollutant gases present in the atmosphere cause green house effect while others do not?

Only gases that can absorb infrared radiation cause the greenhouse effect. Carbon dioxide, methane, water vapour and ozone have molecular structures that vibrate when they absorb infrared rays, so they trap heat. The main gases of the air, nitrogen and oxygen, are simple two-atom molecules that do not absorb infrared radiation, so they let the heat pass through and cause no greenhouse effect, even though they are present in far larger amounts.

4. Descriptive Questions

i. Describe the harmful effects of the major pollutants present in the air.

Carbon monoxide is poisonous because it stops the blood from carrying oxygen. Sulphur dioxide and nitrogen oxides irritate the eyes and lungs, cause asthma and bronchitis, and form acid rain. Particulate matter lodges in the lungs and causes breathing and heart diseases. Ground-level ozone damages the lungs and crops. Lead damages the brain and nervous system, especially in children. Together these pollutants reduce visibility as smog and damage buildings, crops and water life.

4. Descriptive Questions

ii. Explain greenhouse effect. How is global warming dangerous for us?

The greenhouse effect is the warming of the Earth by gases in the atmosphere that let the Sun's short-wave radiation in but absorb the long-wave infrared radiation given off by the Earth, trapping heat like the glass of a greenhouse. Global warming is dangerous because it melts glaciers and polar ice, raising sea levels and flooding coastal areas; it disturbs rainfall, causing droughts in some places and floods in others; it increases heatwaves, storms and diseases; and it destroys habitats, crops and water supplies on which life depends.

4. Descriptive Questions

iii. What is air quality index? What information does it convey?

The air quality index is a number scale used to report how clean or polluted the air is at a place, and what health effects the pollution may cause. A low value means the air is good and healthy, while a high value means the air is unhealthy, especially for sensitive people. It conveys, in one simple number and colour, the levels of pollutants such as particulate matter, ozone, sulphur dioxide and nitrogen dioxide, and it tells people whether it is safe to go outdoors.

4. Descriptive Questions

iv. Who are high risk groups and why is pollution more dangerous for them?

High risk groups are children, elderly people, pregnant women, and people already suffering from asthma, heart disease or other lung diseases. Pollution is more dangerous for them because children's lungs and immune systems are still developing, elderly people's defences are weaker, and people with existing diseases have little reserve to fight further damage. Even pollution levels that healthy adults can tolerate can cause serious illness in these groups.

4. Descriptive Questions

v. Describe three strategies to address environmental issues.

First, reduce pollution at its source: use cleaner fuels, fit filters and catalytic converters, treat industrial waste, and shift to renewable energy such as solar and wind power. Second, protect and restore nature: plant trees, save forests, conserve water, and recycle waste instead of dumping it. Third, make and enforce laws and spread awareness: governments set emission limits, schools teach environmental education, and people use public transport, save energy and avoid single-use plastics.

Chapter 11: Hydrocarbons

In-chapter Exercise (book p. 115)

(1) In the reduction of alkyl halides with Zn / HCl, alkyl halide is being reduced. Which species in this reaction is being oxidized?

Zinc metal is being oxidised. In the reaction Zn + 2HCl gives ZnCl2 + 2[H], zinc loses electrons and its oxidation number rises from 0 to +2, while the alkyl halide gains the nascent hydrogen and is reduced to the alkane. The reducing agent, zinc, is itself oxidised.

In-chapter Exercise (book p. 115)

(2) During the combustion reaction of ethane, which bonds are being broken and which are being formed?

In the combustion of ethane, 2C2H6 + 7O2 gives 4CO2 + 6H2O, the bonds broken are the C-C and C-H bonds of ethane and the O=O double bonds of oxygen molecules. The bonds formed are the C=O double bonds of carbon dioxide and the O-H bonds of water. Breaking bonds takes in energy and forming the new strong bonds releases more energy, which is why combustion gives out heat.

1. Tick the correct answer.

(i) Which other atom is almost always present along with carbon atom in all organic compounds?

(a) Oxygen

(b) Nitrogen

(c) Hydrogen 

(d) Halogen

Correct answer: Hydrogen

1. Tick the correct answer.

(ii) Which other metal can be used to reduce alkyl halides?

(a) Al

(b) Mg ✓

(c) Ni

(d) Co

Correct answer: Mg

Note on Question 1 (ii)

Note: The chapter names only zinc for this reduction.

The chapter's own method for reducing alkyl halides uses zinc with hydrochloric acid, which generates nascent hydrogen. Among the options, magnesium is the other metal that reacts with dilute acid in the same way to generate nascent hydrogen, so Mg is marked. Nickel is used in the chapter only as a catalyst for hydrogenating alkenes and alkynes, not for reducing alkyl halides.

1. Tick the correct answer.

(iii) If naphtha undergoes a combustion reaction what products do you expect it to form?

(a) Alkanes

(b) Alkenes

(c) CO_2 and H_2O ✓

(d) Both alkanes and alkenes

Correct answer: CO_2 and H_2O

1. Tick the correct answer.

(iv) Which alkane will evolve the most amount of heat when it is burnt with oxygen?

(a) Ethane

(b) Propane

(c) n-Butane ✓

(d) iso-Butane

Correct answer: n-Butane

1. Tick the correct answer.

(v) Which reaction is not given by alkanes?

(a) Substitution

(b) Combustion

(c) Addition ✓

(d) Cracking

Correct answer: Addition

1. Tick the correct answer.

(vi) Which hydrocarbon is responsible for explosions in coal mines?

(a) Butane

(b) Pentane

(c) Methane ✓

(d) Ethene

Correct answer: Methane

1. Tick the correct answer.

(vii) Which of the following is not a process of halogenation of alkanes?

(a) Cracking ✓

(b) Chlorination

(c) Bromination

(d) Iodination

Correct answer: Cracking

2. Questions for Short Answers

i. Differentiate between an organic and an inorganic compound.

Organic compounds contain carbon, usually combined with hydrogen, and are generally covalent, with low melting and boiling points, they are often flammable, and their reactions are slow. Inorganic compounds are generally the compounds of metals and non-metals, salts, acids and bases, often ionic, with high melting and boiling points, and their reactions are usually fast. Carbonates, cyanides and oxides of carbon are exceptions and are classed as inorganic.

2. Questions for Short Answers

ii. Why are organic compounds found in large numbers?

Organic compounds are so numerous because of carbon's unique properties. Carbon forms strong bonds with itself, called catenation, giving long chains, branched chains and rings. Carbon is tetravalent and forms single, double and triple bonds with carbon, hydrogen, oxygen, nitrogen and halogens. The same set of atoms can also be arranged in different ways, giving isomers. Together these properties produce millions of different compounds.

2. Questions for Short Answers

iii. Name the products which are obtained when natural gas is oxidized under controlled conditions.

When natural gas, which is mainly methane, is oxidised under controlled conditions with a limited supply of oxygen, methanol (CH_3OH) is obtained. On further controlled oxidation, methanol gives formaldehyde ($HCHO$), and then formic acid. Complete oxidation, by contrast, gives only carbon dioxide and water.

2. Questions for Short Answers

iv. How naphtha fraction is decomposed to give lower hydrocarbons?

Naphtha is decomposed by cracking: it is heated to about 500 C in the presence of a catalyst called zeolite. The large hydrocarbon molecules break apart into smaller, more useful molecules, giving a mixture of alkanes and alkenes containing five to ten carbon atoms. Cracking thus turns the less wanted heavy fractions into petrol-range fuels and chemical feedstocks.

2. Questions for Short Answers

v. Write down the molecular formula, structural formula and the condensed structural formula for iso-butane.

Molecular formula: C4H10. Structural formula: a central CH group joined to three CH3 groups, written as CH3-CH(CH3)-CH3, with the middle carbon bonded to one hydrogen, two CH3 groups in the chain and one CH3 branch. Condensed structural formula: CH3CH(CH3)CH3.

2. Questions for Short Answers

vi. How are organic compounds useful for us?

Organic compounds are useful in almost every part of life. They give us fuels such as petrol, diesel and natural gas; food such as carbohydrates, proteins, fats and vitamins; medicines and drugs; clothes made of cotton, wool and synthetic fibres; plastics, paints, dyes, soaps and detergents; and the paper, rubber and explosives used in industry. Life itself is built of organic compounds.

2. Questions for Short Answers

vii. Write down the names of five such organic compounds which exist naturally?

Five organic compounds found in nature are methane, from natural gas and marshes; glucose, made by plants in photosynthesis; cellulose, the material of

plant cell walls and cotton; starch, stored in potatoes, rice and wheat; and proteins, which build the bodies of animals and plants.

2. Questions for Short Answers

viii. Give IUPAC name to the following compound. CH3-CH(CH3)-CH2-CH3.

Step 1: Find the longest continuous chain: it has four carbon atoms, so the parent is butane. Step 2: Number the chain from the end nearer the branch: the branch falls on carbon 2. Step 3: The branch is a methyl group. The IUPAC name is therefore 2-methylbutane.

2. Questions for Short Answers

ix. How do the melting and boiling points of alkanes change when we move from lower members to higher members?

Melting and boiling points increase as we move from lower to higher members of the alkane series. Larger molecules have more electrons and stronger London dispersion forces between them, so more heat energy is needed to separate them. Methane and ethane are gases, the middle members are liquids, and the higher members are waxy solids.

3. Constructed Response Questions

i. Why do alkanes show little reactivity towards the other reagents?

Alkanes show little reactivity because their C-C and C-H bonds are strong, almost non-polar and fully shared, since the electronegativity values of carbon (2.6) and hydrogen (2.1) differ very little. There is no polar site and no free electron pair for acids, bases or oxidising agents to attack. Alkanes are therefore called paraffins, meaning little affinity, and they react only under vigorous conditions, as in combustion and chlorination.

3. Constructed Response Questions

ii. Why does a mixture of natural gas and air explode?

Natural gas is mainly methane, which burns in air to give carbon dioxide and water with the release of a large amount of heat. When methane is mixed with air in the right proportion and then ignited, every part of the mixture burns at once, and the sudden release of heat expands the gases violently. This rapid combustion is an explosion, and it is the main cause of explosions in homes where gas has leaked and mixed with air.

3. Constructed Response Questions

iii. How do you compare the melting and boiling points of inorganic and organic compounds?

Inorganic compounds generally have high melting and boiling points, because they are often ionic or giant structures held together by strong electrostatic or metallic forces, and a large amount of energy is needed to break them. Organic compounds generally have low melting and boiling points, because they are covalent molecules held together only by weak intermolecular forces, which are easily overcome by heat. Inorganic salts like sodium chloride melt near 800 C, while organic liquids like petrol boil well below 200 C.

4. Descriptive Questions

i. Describe the importance of organic compounds in daily life.

Organic compounds run daily life. Food, the carbohydrates, proteins, fats and vitamins we eat, are organic compounds. Clothes of cotton, wool, silk and synthetic fibres are organic. Medicines, soaps, detergents, cosmetics, paints and dyes are organic. Fuels for cooking, transport and power stations are organic hydrocarbons. Plastics, rubber, paper and furniture materials are organic compounds. Without organic chemistry there would be no food, fuel, medicine or

modern industry.

4. Descriptive Questions

ii. Why is carbon so important as an element that the whole branch of chemistry is based on it?

Carbon is important because it alone combines four properties no other element shares to the same degree. It forms four strong covalent bonds, it bonds to itself in long chains, branches and rings (catenation), it forms single, double and triple bonds, and it bonds with hydrogen, oxygen, nitrogen, sulphur and halogens. These properties produce millions of compounds, including every compound of life, so a whole branch of chemistry, organic chemistry, is devoted to carbon compounds.

4. Descriptive Questions

iii. A carbon-carbon single bond (C - C) in alkanes does not behave as a functional group but a carbon carbon double bond (C = C) in alkenes does. Explain.

A functional group is the part of a molecule that fixes its chemical behaviour, that is, the site where reactions take place. The C-C single bond in alkanes is strong, non-polar and fully shared, so no reagent can attack it, and it does not direct the reactions of the molecule. The C=C double bond in alkenes contains a pi bond, whose electrons are exposed and loosely held, so reagents attack it readily and addition reactions take place there. The double bond therefore behaves as a functional group, while the single bond does not.

4. Descriptive Questions

iv. Explain IUPAC system of nomenclature for alkanes.

In the IUPAC system for alkanes, first find the longest continuous chain of carbon

atoms; its length gives the parent name, using meth- for one, eth- for two, prop- for three, but- for four, pent- for five, and so on, with the suffix -ane. Second, number the chain from the end nearer a branch, so that branches get the lowest possible numbers. Third, name each branch as an alkyl group, such as methyl or ethyl, and write its position number before its name. If two or more identical branches occur, use di-, tri- or tetra-, for example 2,2-dimethylbutane for a butane chain with two methyl groups on carbon 2.

4. Descriptive Questions

v. How combustion reaction of alkanes is useful for us?

The combustion of alkanes is useful because it releases a large amount of heat energy. We burn natural gas and LPG for cooking and heating, petrol and diesel in engines for transport, kerosene in lamps and stoves, and coal, oil and gas in power stations to make electricity. Combustion also provides heat for industry, for example in furnaces and kilns. In every case the useful product is the heat energy released when carbon dioxide and water are formed.

Chapter 12: Laboratory and Practical Skills

1. Tick the correct answer.

(i) Safety in the chemistry laboratory is:

(a) the responsibility of the students only

(b) the responsibility of the professor only

(c) the responsibility of the lab. incharge only

(d) a shared responsibility ✓

Correct answer: a shared responsibility

1. Tick the correct answer.

(ii) Accidents often result from:

(a) making mistakes

(b) failure to use common sense

(c) failure to follow instructions

(d) all of the above ✓

Correct answer: all of the above

1. Tick the correct answer.

(iii) The label Corrosive on a chemical bottle indicates:

(a) that the material is an oxidizing agent

(b) that the material can degrade rapidly upon exposure

(c) that its contact destroys living tissue ✓

(d) that the chemical can explode

Correct answer: that its contact destroys living tissue

1. Tick the correct answer.

(iv) When diluting an acid with water:

(a) do it quickly

(b) do not stir the container

(c) always add acid to water

(d) always add water to acid

Correct answer: always add acid to water

1. Tick the correct answer.

(v) What should you do in case of a fire drill in the lab.?

(a) run to safety shower

(b) climb into the fume cupboard

(c) close gas valves and turn off all equipments

(d) carry chemicals out of the lab.

Correct answer: close gas valves and turn off all equipments

2. Questions for Short Answers

i. Name some corrosive chemicals.

Common corrosive chemicals are concentrated sulphuric acid, concentrated nitric acid, concentrated hydrochloric acid, sodium hydroxide (caustic soda), potassium hydroxide (caustic potash), and strong bleaching solutions. All of them destroy skin and other living tissue on contact and damage eyes, clothes and bench surfaces.

2. Questions for Short Answers

ii. What type of safety precautions are adopted to avoid damage due to explosive chemicals?

Explosive chemicals are stored in small amounts in cool, dry places away from flames, heat, shocks and friction. They are never heated directly, never shaken or dropped, and are handled gently with protective equipment. Containers are kept tightly closed and labelled, and only the small amount needed is taken to

the bench at a time.

2. Questions for Short Answers

iii. Why signs and symbols are posted on lab. and chemical bottles?

Signs and symbols are posted to warn everyone at a glance about the dangers inside a laboratory or a bottle, even before reading any text. They show whether a substance is corrosive, flammable, toxic or explosive, and they guide safe handling, storage and first aid. Clear labelling prevents mistakes, such as using the wrong chemical, and helps people act correctly in an emergency.

2. Questions for Short Answers

iv. How fire caused by chemicals should be handled?

A chemical fire should be handled by first raising the alarm and switching off gas and electricity if it is safe to do so. The right extinguisher must be used: a carbon dioxide or dry powder extinguisher for most chemical and electrical fires, never water on burning oil, solvents or live equipment. If the fire is large, everyone must evacuate at once and the fire service must be called; no one should risk their life to save equipment.

2. Questions for Short Answers

v. Why emergency drills are important to face emergency situations?

Emergency drills are important because they train students and staff to act quickly and correctly instead of panicking when a real emergency happens. In drills people practise evacuating, using extinguishers, closing gas valves and helping the injured, and the equipment is checked to see that it works. A practised team saves lives and property that an untrained group would lose.

3. Constructed Response Questions

i. How will you handle an emergency situation caused by fire due to short circuiting?

First, stay calm and raise the alarm so that everyone evacuates. Switch off the main power supply if it can be done safely, because water or foam on a live electrical fire is deadly. Use a carbon dioxide or dry powder extinguisher on the fire, never water. Close doors to confine the fire, keep away from smoke, and call the fire service and assist the emergency staff when they arrive.

3. Constructed Response Questions

ii. What type of reactions should be carried out in fume cupboard?

Reactions that release toxic, irritating or foul-smelling gases and vapours must be carried out in a fume cupboard. Examples are reactions giving off chlorine, sulphur dioxide, nitrogen dioxide, ammonia or hydrogen chloride gas, reactions with volatile acids, and any work with toxic or strongly smelling chemicals. The fume cupboard draws the harmful vapours away from the worker and out of the laboratory.

3. Constructed Response Questions

iii. Why sudden shock can cause some chemicals to explode?

Some chemicals, such as explosives and certain peroxides, are unstable and need only a small amount of activation energy to start breaking down. A sudden shock, a knock, friction or a sharp blow supplies that energy all at once, and the decomposition then releases a large amount of heat and gas in a fraction of a second. The rapid release of gas and heat is the explosion, which is why such chemicals must never be shaken, dropped or struck.

4. Descriptive Questions

i. Explain hazards due to explosive and toxic chemicals.

Explosive chemicals can detonate on heating, shock or friction, causing fire, blast, flying glass and serious injuries, and they can destroy the laboratory. Toxic chemicals poison the body when breathed in, swallowed or absorbed through the skin: they can damage the lungs, liver, kidneys and nervous system, and some cause cancer or unconsciousness. Both kinds of chemicals must be stored safely, used in small amounts, and handled only with protective equipment and proper ventilation.

4. Descriptive Questions

ii. Write down five such common safety instructions which are used to avoid all types of hazards.

First, always wear safety goggles, a lab coat and gloves, and tie back long hair. Second, never taste a chemical and never smell it directly; waft the vapour gently towards the nose if the teacher asks. Third, always add acid to water slowly, never water to acid, when diluting. Fourth, read the label of every bottle before use, and never use an unlabelled chemical. Fifth, know the emergency exits, the position of the fire extinguisher, the safety shower and the first aid box, and report every accident to the teacher at once.

4. Descriptive Questions

iii. Explain the importance of warning signs and symbols to avoid any accident in the lab.

Warning signs and symbols are important because they give instant information about danger without needing words or a long explanation. A student who sees a flame symbol knows the substance is flammable, a skull warns of poison, and a corrosive symbol warns that contact destroys skin. This instant warning stops careless handling, reminds workers of the right precautions, and in an emergency it helps rescuers know at once what kind of danger they face, so

accidents are prevented before they happen.

4. Descriptive Questions

iv. Name some toxic chemicals. Describe the effects of spreading toxic gas in the lab.

Some toxic chemicals are chlorine, ammonia, hydrogen sulphide, carbon monoxide, sulphur dioxide, mercury and cyanide compounds. If a toxic gas spreads in the laboratory, it is breathed into the lungs and can cause coughing, choking, headache, dizziness and unconsciousness, and in high amounts it can kill. Eyes and skin are also irritated. Everyone must leave the room at once, doors should be closed to confine the gas, ventilation and the fume system should be used, and the teacher and emergency staff must be informed immediately.

4. Descriptive Questions

v. A student has spilled over a corrosive and explosive chemical due to an accident. Which emergency measures you will take to tackle the situation?

First, stay calm, raise the alarm and evacuate everyone from the area, keeping all flames, sparks and electrical equipment away because the chemical is explosive. Remove the student's contaminated clothes at once and wash the affected skin with plenty of running water for at least fifteen minutes. Do not rub the skin and do not try to neutralise the chemical on the body. Call the teacher and the emergency staff, keep the student under observation, and let trained staff clean the spill with the correct absorbent and protective equipment.